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ISRO

A glorious space story

In 1981, the Indian Space Research Organisation (ISRO) carried its APPLE Satellite on a bullock cart. In four decades, India has achieved global leadership, with ISRO launching 129 satellites of Indian origin and 342 foreign satellites from 36 countries. This satellite-enabled data and services capability acted as building blocks of India's IT and telecom revolution, offering a platform for game-changing solutions– from weather forecasting to television broadcasting to mobile communications. Its impact traverses multiple dimensions. For instance, the National Stock Exchange (NSE) did satellite-based financial market trading for the first time in the world. VSAT technology was used to build computer-based order matching at the exchange, resulting in a quantum leap in the quality of the Indian equity market.

GETTY IMAGES



Covid Vaccination

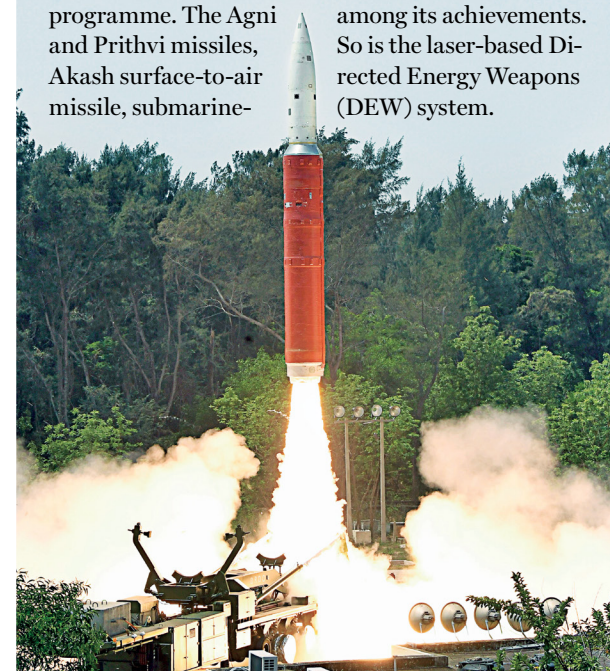
India as the 'Vaccine Capital of the World'

India had already established its reputation as a leader in research, innovation, manufacture and deployment of vaccines, be it polio, hepatitis B, rotavirus, etc. But this Indian capacity was stress-tested during the Covid-19 pandemic. The Serum Institute of India came out with a Covid vaccine based on licensed technology. The National Institute of Virology and Bharat Biotech jointly developed Covaxin. These indigenous vaccines saved millions of lives. The effective management and administration of over two billion doses within a record time of 18 months was made possible by the CoWIN platform, built by the National Informatics Centre.

DRDO

From tech-deprivation to techno-nationalism

Despite the denial of strategic technologies, the Defence Research and Development Organisation (DRDO) achieved many milestones, such as the rocket and missile programme. The Agni and Prithvi missiles, Akash surface-to-air missile, submarine-launched ballistic K-15 missile system and supersonic cruise missile BrahMos, which was developed in partnership with Russia and was the fastest in its category when introduced, are among its achievements. So is the laser-based Directed Energy Weapons (DEW) system.



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